Code: 20A04604b **R20**

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

EMBEDDED SYSTEM DESIGN

(Electronics and Communication Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

(a)	What is an embedded system?	2M
(b)	What is the purpose of embedded system?	2M
(c)	What is the role of Real Time Clock (RTC) in embedded system?	2M
(d)	What is the difference between PLD and ASIC?	2M
(e)	Write the differences between synchronous and asynchronous communication interfaces.	2M
(f)	What are the advantages and disadvantages of SPI?	2M
(g)	What is macro in ‘Embedded C’ programming?	2M
(h)	What is the difference between ‘recursion’ and ‘iteration’?	2M
(i)	What is Priority Inversion?	2M
(j)	What is binary semaphore? Where is it used?	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	(a)	Discuss the features and architectural considerations pertinent to embedded systems.	5M
	(b)	Provide the classification of embedded systems based on generation and complexity.	5M
		OR	
3	(a)	Explain the significance and necessity of system integration in embedded systems.	5M
	(b)	Write a note on history of embedded systems.	5M
4	(a)	Explain different types of memories.	5M
	(b)	What are the different types of embedded processors? Explain.	5M
		OR	
5	(a)	Explain the role of Watchdog Timer in embedded system.	5M
	(b)	What is sensor? Explain its role in embedded system design. Illustrate with an example.	5M
6	(a)	Explain the different external communication interfaces in brief.	5M
	(b)	Compare the operation of Wi-Fi and ZigBee networks.	5M
		OR	
7	(a)	Explain the RS-232 C serial interface in detail.	5M
	(b)	Distinguish between I2C and SPI communication interface.	5M
8	(a)	Explain the various steps involved in the assembling of an assembly language program.	5M
	(b)	What is ‘pointer’ in embedded C programming? Explain its role in embedded application development.	5M
		OR	
9	(a)	What is Assembly language Programming? Explain the format of assembly language instruction.	5M
	(b)	Explain ‘structure’ in the embedded C programming context.	5M
10	(a)	What is task control block (TCB)? Explain the structure of TCB.	5M
	(b)	Explain ‘tasks, process and threads’ in the operating system context.	5M
		OR	
11	(a)	Explain multiprocessing, multi-tasking, and multiprogramming.	5M
	(b)	What is deadlock? What are the different conditions favoring deadlock. Also mention different methods of handling deadlocks.	5M

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